

Positional Semantics for Retrieved Native-KV Memory: RoPE, Fragmentation, and Bounded Relocation in Progressive Retrieval Attention

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Abstract

Retrieved native-key/value (K/V) memory reuses attention states that were encoded outside the current forward pass. This raises two questions that ordinary prefix caching can avoid: which source position does each cached key represent, and which causal context shaped it? We give Progressive Retrieval Attention (PRA) an explicit positional contract while preserving learned absolute position embeddings as the default and adding rotary position embedding (RoPE) to direct and retrieved-memory attention. In five-seed controlled tiny and small models, a compact logical start offset per contiguous chunk removes first-layer position-reset error exactly for both positional families. It does not remove deeper contextualization error; 50% overlap helps three of four tier–position combinations but slightly worsens the small absolute mean. A four-cell pre/post-RoPE storage matrix establishes exact output parity when effective positions match and shows that K-only rebinding intentionally changes attention. Deferred rotation is therefore a flexibility mechanism, not an accuracy improvement, and raises the unfused warm CUDA path from 0.179 to 0.649 ms in the measured small configuration. A continuously advancing implicit-head experiment keeps every model operation at or below 24 tokens while the logical source spans 192 tokens. Logical offsets and overlap often lower answer loss, but a constructed single-chunk oracle is not uniformly better for tiny models, exposing a separate selection and context-composition problem. These results separate positional identity, contextual continuity, and retrieval quality. They validate the transport contract, not production long-context quality.

1 Introduction

A transformer key is not only a content representation. Depending on the positional mechanism, it carries or interacts with a coordinate. This is easy to ignore while every key is produced in one dense forward pass. It becomes operational when a system encodes a long source in bounded blocks, stores native per-layer K/V, retrieves a sparse subset, and combines it with a later query. A source key may retain its document coordinate, restart at zero in each block, or receive a new coordinate when retrieved. These choices are not equivalent.

PRA provides a controlled setting for studying the problem. It stores full layer-native K/V behind inexpensive routing gists and inserts selected states before direct K/V under one softmax. Paper 1 established native transport, exact routing, model-bounded encoding, and an implicit prompt-head reference. The present paper holds those mechanisms fixed and changes the positional semantics.

This paper asks a narrower question than whether a model can extrapolate to a longer dense context: can independently produced native K/V retain an auditable position identity, and what

errors remain after that identity is repaired? Our contributions are:

- a reusable position strategy supporting learned absolute, sinusoidal, and RoPE semantics without changing the default absolute checkpoint layout;
- a weight-preserving RoPE SelfAttention-to-PRA conversion whose no-memory logits match the source model;
- compact source-relative offsets, explicit position metadata, and pre/post-position state labels on cached native K/V;
- probes separating position reset, contextual block fragmentation, overlap, pre- versus post-RoPE K, retrieved-distance policies, and continuously advancing implicit-head behavior; and
- five-seed tiny/small evidence, including negative routing and variance results that constrain the claims.

2 Progressive Retrieval Attention in Brief

PRA keeps a recent direct tail continuously visible and makes older or external material addressable. A long source is first encoded in model-safe blocks. Each layer publishes its native K/V, and those states are sliced into smaller routing chunks. A gist is a compact routing summary of a chunk; it is *not* the chunk’s K/V and is never used as a substitute attention value. At query time, gist scores select chunks, the selected chunks’ full token-level native K/V are materialized, and memory K/V attend beside direct K/V under one softmax.

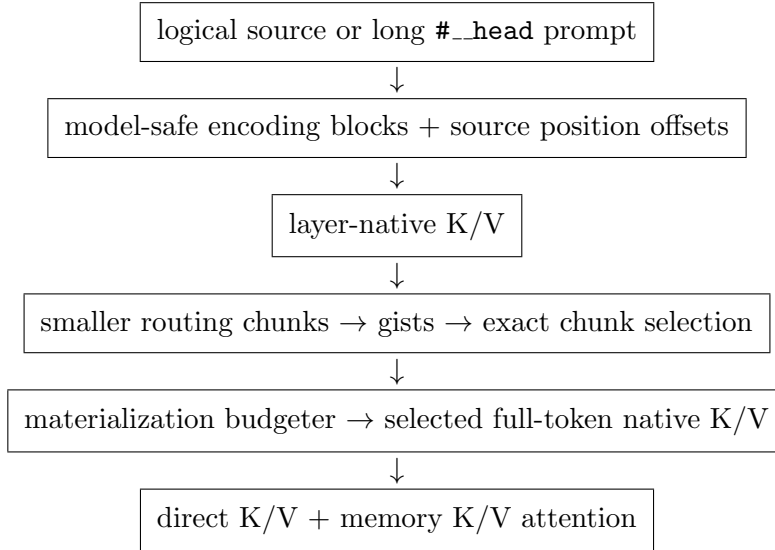


Figure 1: PRA data path. Gists choose chunks; attention consumes the selected chunks’ native token-level K/V. Position continuity is established during bounded encoding and audited again at materialization.

Four lengths describe different constraints:

$$L_{\text{logical}} \gg L_{\text{model_op}} \geq L_{\text{encode}} > L_{\text{route}}, \quad L_{\text{direct}} + L_{\text{materialized}} + L_{\text{reserve}} \leq L_{\text{model_op}}. \quad (1)$$

L_{logical} is the source coordinate span. $L_{\text{model_op}}$ is the maximum number of tokens in one model invocation. L_{position} , introduced below, is the largest representable coordinate and need not equal the operation limit. L_{encode} bounds contextual cache construction, while L_{route} controls retrieval granularity. A long logical source therefore does not imply a long dense model call.

2.1 A concrete position-reset failure

Consider tokens A–H at logical positions 0–7, encoded as A–D and E–H. If the second block restarts at zero, E receives the coordinate formerly assigned to A. Source-relative encoding instead uses

$$p_{\text{logical}}(j) = p_{\text{chunk, start}} + j. \quad (2)$$

Source token	A	B	C	D	E	F	G	H
Source position	0	1	2	3	4	5	6	7
Reset chunk position	0	1	2	3	0	1	2	3
Offset chunk position	0	1	2	3	4	5	6	7

Thus token 37 within a block beginning at 4096 receives position 4133. A contiguous routing chunk needs one integer start offset, not a position tensor per token; explicit tensors remain useful for irregular layouts and audit output.

For a continuously advancing prompt head, rollover preserves one coordinate system as direct tokens become historical memory. Independent URI references are different: each source owns its own coordinate system unless an application defines a shared layout. The implementation therefore treats source identity and source-relative offset as cache provenance rather than silently concatenating unrelated documents into one global axis.

3 Background and Related Work

Learned absolute embeddings add a position-dependent vector before the decoder stack and therefore entangle content and position in later hidden states. Transformer-XL introduced relative positional treatment for recurrent segment memory [2]. Shortformer moved absolute position information to queries and keys to improve cached-state reuse across subsequences [6]. ALiBi adds distance-dependent attention biases for train-short, test-long extrapolation [7].

RoPE rotates query and key feature pairs by position-dependent phases, making their dot product depend on relative displacement while each vector retains an absolute phase [8]. Position Interpolation rescales indices into a trained range [1]; LongRoPE uses nonuniform interpolation and progressive extension [3]. Those methods enlarge dense model context. Our question is different: how should positions behave when only selected externally stored native K/V is returned? StreamingLLM preserves selected chronological cache states and attention sinks [9]; PRA retrieves URI- or request-addressed chunks, so it must also specify source positions, relocation, and routing semantics.

Cache reuse systems expose adjacent parts of the same problem. Prompt Cache reuses modular attention states while preserving a position-aware prompt schema [4]. CacheBlend observes that independently cached non-prefix chunks miss preceding context and selectively recomputes states [11]. MiniPIC stores unrotated keys and binds RoPE positions inside attention [5]; SemPIC instead learns a cache writer to reduce residual context dependence [10]. These approaches motivate the same distinction tested here: correcting coordinates does not reconstruct absent causal context, and flexible position binding does not by itself select useful memory.

4 Positional Semantics for Native K/V

4.1 What RoPE invariance does and does not say

For one head of width d_h , let R_p be the block-diagonal rotation at position p :

$$\tilde{q}_i = R_i q_i, \quad \tilde{k}_j = R_j k_j. \quad (3)$$

Orthogonality and the group property give

$$\tilde{q}_i^\top \tilde{k}_j = q_i^\top R_i^\top R_j k_j = q_i^\top R_{j-i} k_j. \quad (4)$$

Thus translating both positions by c leaves the score unchanged: $(R_{i+c} q_i)^\top (R_{j+c} k_j) = q_i^\top R_{j-i} k_j$. This supports relocation of a complete query–memory relation. It does not imply that moving only a cached key leaves attention unchanged; a fixed query should observe the changed relative distance.

4.2 Three independent choices

Three choices are often collapsed into "RoPE support":

1. **Encoding positions:** local block indices or logical source indices.
2. **Storage state:** pre-positional K_{raw} plus metadata, or published post-RoPE K_{rope} .
3. **Materialization policy:** preserve source distance, reset locally, relocate as a block, compress distance, or retain order only.

The canonical implementation publishes post-position K. Pre-position storage remains experimental because deferred rotation changes the cache representation and runtime work.

4.3 Position and contextualization fragmentation

If a dense source is split into bounded blocks, local position reset changes absolute embeddings and post-RoPE key phases. Independently, later blocks lose causal left context, which changes hidden states even with perfect position IDs. We call these *positional fragmentation* and *contextualization fragmentation*. Logical global positions can remove the first at layer 0. Encoding overlap can only approximate the second.

Table 1: Failure modes that must be measured separately.

Failure	Cause	Diagnostic	Mitigation/control
Positional fragmentation	Chunk coordinates reset	Layer-0 K mismatch	Source-relative offsets
Contextualization fragmentation	Earlier causal context is absent	Deeper mismatch after offset repair	Overlap or recomputation
Selection failure	Router misses useful chunks	Routed–oracle gap and gist recall	Better routing
Positional rigidity	Phase is baked into K	Deliberate rebinding probe	Raw K plus deferred RoPE
Remote approximation	Relative phase is compressed	Exact-distance output deviation	Policy-specific adaptation

5 Implementation

The shared code adds a registry-backed `PositionEncoding` interface. Its first method positions token embeddings; its second transforms per-head Q/K with shape $[B, H, T, d_h]$. Absolute mode adds the existing learned table and leaves projected Q/K unchanged. RoPE mode leaves token embeddings unchanged and rotates Q/K. A position-aware SelfAttention block is instantiated only for non-absolute modes, so old absolute models retain their original `TransformerEncoderLayer` state names and behavior.

Table 2: Position mechanisms under retrieved native-K/V reuse.

Mechanism	Source offsets	Coordinate range	Retrieval-time re-binding
Learned absolute	Added before layer 0	Limited by learned table L_{position}	Not separable from hidden state
Sinusoidal absolute	Added before layer 0	Formula-defined; extrapolation unverified	Not separable from hidden state
RoPE, post-position K	Applied to Q/K	Formula-defined; extrapolation unverified	No, without inverse rotation
RoPE, pre-position K	Applied when consumed	Formula-defined; extrapolation unverified	Yes, by choosing effective IDs

The configuration separates `model_max_context_tokens`, the native operation limit, from `position_capacity`. Learned absolute IDs are checked against the latter. This allows, for example, a 32-token bounded operation to use positions 160–191 in a model with a 192-entry position table. RoPE accepts larger integer IDs, but acceptance by the formula is not evidence that a model trained only on short distances will use them well.

PRA attention applies the same strategy in direct and cache-publication paths:

```

q = split_heads(q_proj(hidden))      # [B,H,T,Dh]
k = split_heads(k_proj(hidden))      # [B,H,T,Dh]
q, k = position.transform_qk(q, k, position_ids)

k_ref = split_heads(k_proj(reference_hidden))
_, k_ref = position.transform_qk(k_ref, k_ref, reference_positions)
cache.publish(k_ref, v_ref, position_ids, state="post_position")

```

Values are not rotated. Each `LayerKV` records the positions used to publish K and a `position_state`. `ReferenceChunk` and `ReferenceChunkMemory` carry `logical_start` and `logical_end`; validation, overlap, overflow splitting, routing selection, CPU residency, and materialization preserve them. A contiguous chunk can reconstruct token IDs as `logical_start + arange(length)`. The result trace exposes the same range so a consumer can audit which native states entered attention.

Conversion copies Q/K/V projections, output projection, norms, feed-forward layers, embeddings, and language-model head from trained RoPE SelfAttention into PRA. With memory disabled, source and converted logits match to unit-test tolerance. No memory adapter or additional transport projection is introduced.

For a long prompt, the adapter keeps a direct tail and registers displaced tokens under `#_head`. Bounded encoding produces native K/V, smaller chunks produce routing gists, and whole chunks

are selected under

$$T_{\text{direct}} + T_{\text{memory}} + T_{\text{reserve}} \leq L_{\text{model_op}}. \quad (5)$$

RoPE can continue logical IDs past $L_{\text{model_op}}$ without enlarging an operation. Learned absolute mode can do the same only while IDs remain below L_{position} .

6 Experimental Design

6.1 Models and training

Controlled tiers are smaller than repository deployment profiles so all paired seeds fit one NVIDIA GTX 950M. Both modes retain the learned position table, unused by RoPE, to match parameter count exactly.

Table 3: Controlled matched model tiers.

Tier	Layers	Heads	d	d_{ff}	Parameters	Steps
Tiny	2	4	64	128	92,288	100
Small	4	4	128	256	580,608	150

Seeds are $\{1, 7, 21, 42, 87\}$. Each absolute/RoPE pair uses the same 2,048 fixed-target synthetic training examples, optimizer, batches, vocabulary, 192-token native context, and training tokens. Every run reaches 100% held-out answer-token accuracy. The downstream set has 12 invariant source/query/answer examples; only source boundaries change. We evaluate 2, 5, 16, 32, and 64 splits.

PRA uses post-position K, local positions per independent reference, a 160-token whole-chunk budget, and a four-token reserve in the split experiment. Conditions include dense full context, tail only, native all, evidence oracle, gist-routed, shuffled, and memory-disabled. Recovered context benefit is

$$\text{RCB} = \frac{\mathcal{L}_{\text{tail}} - \mathcal{L}_{\text{PRA}}}{\mathcal{L}_{\text{tail}} - \mathcal{L}_{\text{full}}}, \quad (6)$$

reported without clipping.

6.2 Additional probes

The pure probe fixes Q/K/V across shifts up to 8,192 and measures logit RMSE, Jensen–Shannon divergence, top-token agreement, and output RMSE. The pre/post probe compares stored post-RoPE K with deferred rotation at identical positions, K-only relocation against a fixed query, and common Q/K translation.

The raw-K policy probe holds Q/K/V content fixed while source–query distance grows from 1L to 32L, where $L = 192$. Exact logical, immediately preceding local, 192-token clipped, log-compressed, four-level bucketed, and remote-past assignments move each 32-token chunk as a unit, preserving local order. This is a positional sensitivity probe, not downstream quality. For systems cost, synchronized warm CUDA measurements separately time selected-key gather or gather-plus-rotation, attention, and their combined path over 500 iterations for each of five seeds.

The source-continuity decomposition compares dense 128-token encoding with 32-token blocks under local reset, source-relative offsets, and 25% or 50% left overlap. It reports K RMSE by layer and paired seed. The absolute table has 192 entries, so all logical IDs are representable even though each encoding operation is bounded to 32 tokens.

The RoPE storage matrix crosses post- versus pre-position K with reset versus source-relative coordinates: A is post/reset, B post/offset, C pre/reset, and D pre/offset. B is the reference. A separate raw-K condition rebinds the source by +512. Runtime records gather, compact position reconstruction, RoPE transform, materialization, attention, and the warm combined path. All timings synchronize CUDA around 500 iterations and use five seeds.

The continuous-head probe uses $L_{\text{logical}} = L_{\text{position}} = 192$ and $L_{\text{model_op}} = 32$. It keeps 8 direct tokens, encodes 16-token blocks, routes 8-token chunks, and materializes at most 24 memory tokens. Four examples and five seeds progress from reset routing to offset routing, offset plus 50% overlap, and a constructed oracle containing the first evidence chunk. This oracle tests one selection hypothesis; it is not ground-truth proof that no other chunk helps. The earlier distance-scaling head probe is retained separately.

Before result inspection we recorded the following expected directions: offsets should reduce reset error; overlap should reduce the remaining contextual error; matched pre/post storage should agree; intentional K-only rebinding should change attention; and the constructed oracle should be no worse than routed selection. We report failed and partial expectations rather than discarding them.

7 Results

7.1 Pure translation and storage

Across head dimensions 8–64, common translation preserves the top-attended token exactly. Maximum logit RMSE is 5.79×10^{-5} , maximum JS divergence is 3.00×10^{-7} , and maximum output RMSE is 1.80×10^{-5} (Figure 2).

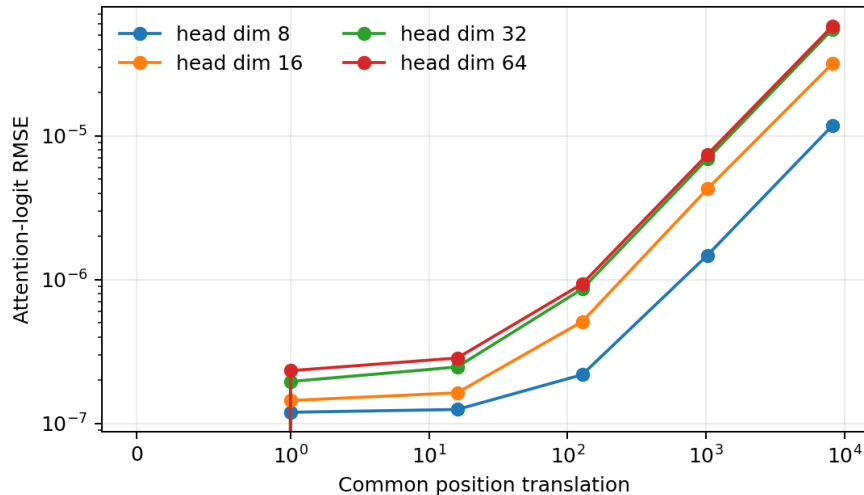


Figure 2: Numerical RoPE common-translation invariance. The logarithmic vertical axis exposes floating-point error, not a quality effect.

The trained-model storage matrix gives the same result at the model output (Figure 3). Post/reset A equals pre/reset C exactly; post/offset B equals pre/offset D exactly. B and D reproduce the offset reference with zero output RMSE and top-token agreement 1. Reset A/C instead have mean K RMSE 0.648, output RMSE 0.063, and top-token agreement 0.225. Rebinding

raw K by +512 gives output RMSE 0.093 and agreement 0.05. The matrix rejects two possible misconceptions: raw K is not intrinsically more accurate when effective IDs match, and changing K coordinates alone is not an invariant operation.

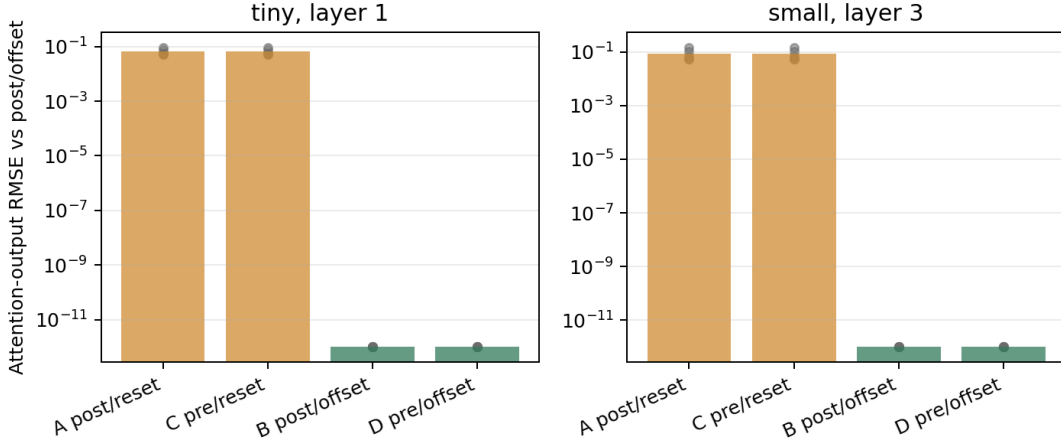


Figure 3: Four-cell RoPE storage matrix over both tiers, all layers, and five seeds. Deferred rotation is exact at matched effective positions; reset coordinates and intentional rebinding produce the expected attention changes.

The earlier tensor probe agrees: moving K alone by 128–8,192 positions changes fixed-query scores, while translating Q and K together remains near invariant (maximum RMSE 4.91×10^{-4} at 8,192). Storage state therefore controls when positions can be bound, not whether the same binding is accurate.

7.2 Deferred rotation and distance precision

Exact logical assignment reproduces post-position K, scores, and outputs. The approximate policies are not numerically neutral. At 32L in the small tensor tier, their mean output RMSE against exact assignment spans 0.289–0.326 and top-token agreement is 0–0.05. Tiny-tier output RMSE spans 0.259–0.290 and agreement is 0.05–0.15. These are five-seed random-content attention probes: they show geometric sensitivity, not task-loss degradation. A compressed policy therefore requires training or adaptation evidence before deployment.

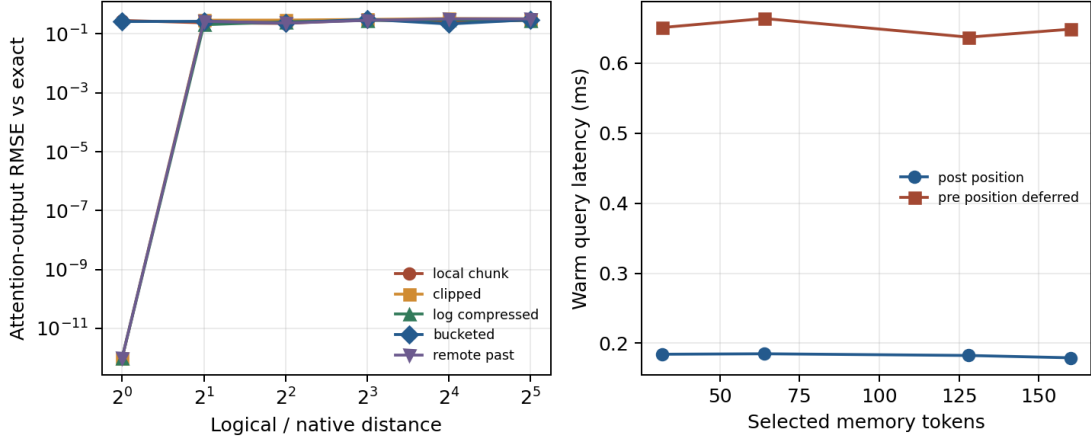


Figure 4: Left: positional-policy deviation from exact logical assignment for the small tensor tier. Right: synchronized warm CUDA microbenchmark. Approximate distance is not a no-op, and deferred rotation adds a launch-dominated operation on this GPU.

Table 4: Small-tier CUDA timing in ms, mean over five seeded runs of 500 warm iterations.

Storage	K/V	Gather	Reconstruct	RoPE	Attention	Warm
Post-position	32	.0312	.0000	.0000	.1343	.1842
Pre-position + deferred	32	.0337	.0191	.4021	.1343	.6511
Post-position	160	.0327	.0000	.0000	.1342	.1791
Pre-position + deferred	160	.0329	.0145	.3838	.1342	.6487

For 192 stored small-tier tokens, both modes use 196,608 bytes of fp32 K/V. The auditable cache can reconstruct a contiguous position range from an 8-byte start offset per chunk; a full 192-token int64 audit tensor would cost 1,536 bytes (0.781%). Selecting 160 tokens materializes 163,840 K/V bytes. Raw K changes when phase can be assigned, not how many K/V elements are retained. Position reconstruction is small; the unfused RoPE transform dominates prototype overhead. The nearly flat token-count timing reflects kernel-launch dominance on a GTX 950M, so this experiment supports no production throughput claim.

7.3 Split scaling

Figure 5 reports the trained experiment. Split 2 often creates a chunk too large for the 160-token whole-chunk budget, so native-all becomes tail-only. This is a materialization outcome, not a positional failure.

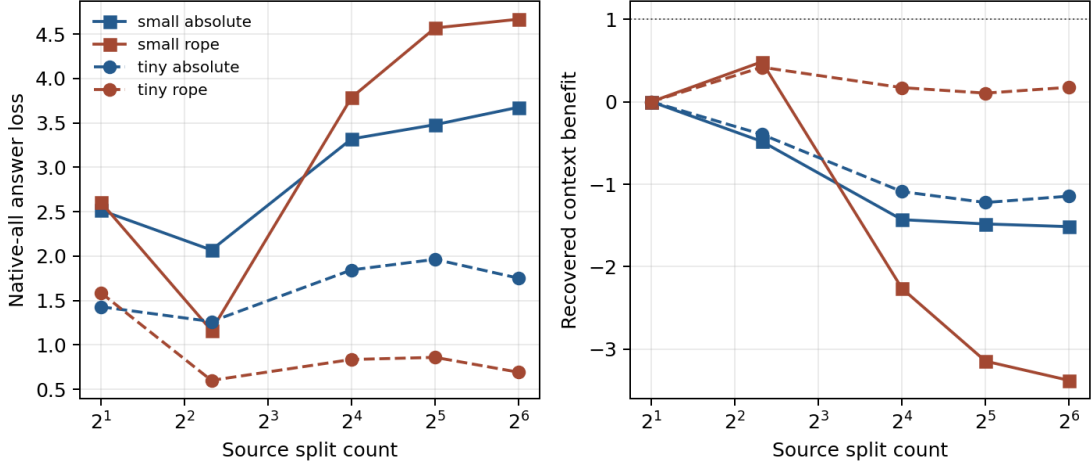


Figure 5: Five-seed native-all loss and recovered context benefit. Negative RCB means that unfiltered memory hurts relative to tail-only inference.

Table 5: Mean native-all loss by split count. Lower is better.

Tier	Position	5	16	32	64
Tiny	Absolute	1.264	1.845	1.964	1.752
Tiny	RoPE	0.601	0.837	0.860	0.693
Small	Absolute	2.069	3.321	3.481	3.677
Small	RoPE	1.163	3.787	4.570	4.669

Tiny RoPE is lower in all five paired seeds at 16 and 32 splits. With five seeds, however, the smallest exact two-sided sign probability is 0.0625. At 64 splits it is lower in four of five. The small tier reverses in mean at 16–64 and has high seed variance. There is no scale-independent RoPE advantage.

Oracle selection isolates transport and sparsity from routing. At 64 splits, tiny absolute/RoPE oracle losses are 0.581/0.070, corresponding to RCB 0.483/0.949; RoPE is lower in all five seeds. Small oracle losses are both low, 0.126/0.046, with RCB 0.967/0.929. Routed losses remain much larger (tiny 1.994/1.301; small 4.434/4.721). Strong oracle outcomes beside weak routed outcomes locate much of the remaining failure in selection, not transport.

7.4 Source-relative continuity and overlap

Source-relative offsets make layer-0 K exactly match the dense 128-token comparator in every tier, positional family, and seed. Reset layer-0 K RMSE ranges from 0.559 to 0.667 by tier–position mean. The exact repair under both learned absolute embeddings and RoPE shows that source coordinate identity, rather than RoPE specifically, is the necessary first step.

Deeper layers retain contextualization fragmentation (Table 6). Offsets lower final-layer RMSE in all five seeds for all four model conditions. Fifty-percent overlap then lowers every tiny seed, lowers four of five small RoPE seeds, but lowers only two of five small absolute seeds and slightly worsens that mean. It processes 1.375 times as many tokens. Overlap is therefore a tunable historical-context approximation, not a guaranteed or monotonic repair.

Table 6: Final-layer K RMSE from dense encoding. Cells are mean / median / standard deviation over five seeds. Lower is better.

Tier	Position	Reset	Offset	Offset + 50%
Tiny	Absolute	.6769/.6860/.0339	.1129/.1064/.0214	.0670/.0625/.0150
Tiny	RoPE	.6745/.6649/.0393	.1373/.1397/.0362	.0690/.0615/.0213
Small	Absolute	.7193/.6694/.0953	.3066/.2870/.0930	.3103/.2959/.1634
Small	RoPE	.7498/.7234/.0899	.4235/.4198/.1512	.2990/.2378/.1212

The 25% overlap means, shown in Figure 6, are 0.0849, 0.0966, 0.2918, and 0.2806 in table order.

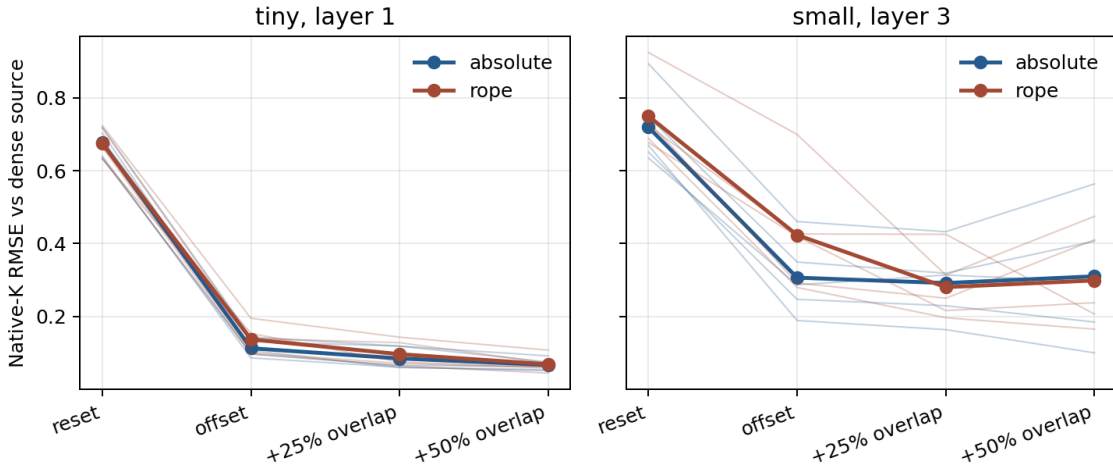


Figure 6: Position-reset decomposition across all paired seeds. Offsets remove layer-0 error; the plotted final-layer progression shows the remaining context dependence and the nonuniform effect of overlap.

7.5 Continuous implicit-head progression

The new head probe advances one logical coordinate system through rollover. Every model call contains at most 24 tokens, below $L_{\text{model.op}} = 32$, while source positions span $L_{\text{logical}} = 192$; native-limit violations are zero. Table 7 shows that representation repair does not imply uniform end-task behavior.

Table 7: Continuous-head answer loss. Cells are mean / median over four examples and five seeds.

Tier	Position	Reset routed	Offset routed	Offset + overlap	Constructed oracle
Tiny	Absolute	2.3837/2.0986	.7304/.4736	.6474/.2631	.1384/.1136
Tiny	RoPE	2.5910/1.5186	1.5547/.9828	.8763/.1163	1.0124/.6812
Small	Absolute	3.8306/2.2442	.6681/.0587	.0030/.0027	.0022/.0020
Small	RoPE	6.7887/9.2938	6.2249/8.5253	3.0396/1.7279	.0012/.0011

Offsets lower seed-mean loss in 5/5 tiny absolute, small absolute, and small RoPE seeds, but only 3/5 tiny RoPE seeds. Overlap lowers all tiny RoPE and small RoPE seeds, 4/5 small absolute

seeds, and 2/5 tiny absolute seeds. Evidence-chunk recall rises from reset to offset/overlap for all four conditions, but remains only 0.312 for small RoPE before the oracle. The oracle sets recall to one and nearly eliminates small-model loss. It is not universally superior: tiny RoPE mean loss worsens from 0.8763 to 1.0124 and only two seeds improve. A single selected evidence chunk can remove useful surrounding context, so this is a failed preregistered expectation rather than evidence against the position contract. Across reset, offset, overlap, and oracle, the corresponding loss standard deviations are 1.9282/.7839/.8139/.0979 for tiny absolute, 2.3486/1.1528/1.1969/1.1443 for tiny RoPE, 3.7387/1.1183/.0012/.0009 for small absolute, and 3.7760/3.7089/2.6464/.0002 for small RoPE.

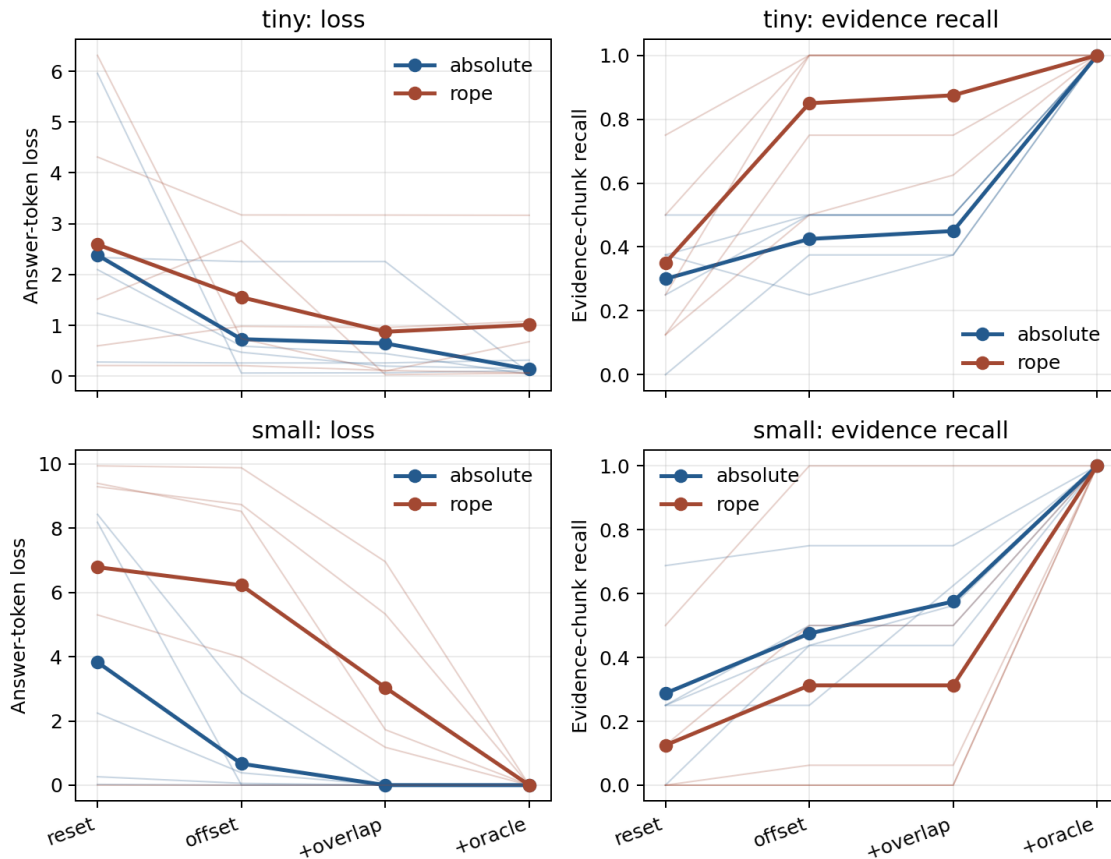


Figure 7: Continuous-head loss and evidence-chunk recall. Paired seed traces expose variation hidden by means; the separately recorded operation bound is independent of logical span.

Table 8: Recorded expectation versus observation. “Partial” means the mean or most seeds move as expected, but not all paired seeds.

Expectation	Tiny absolute/RoPE	Small absolute/RoPE	Outcome
Offsets reduce representation error	Yes / Yes	Yes / Yes	Supported
50% overlap further reduces representation error	Yes / Yes	No / Partial	Mixed
Matched pre/post RoPE storage agrees	Exact for every tier/layer/seed		Supported
K-only rebinding changes attention	Mean output RMSE .0931		Supported
Offsets lower continuous-head loss	Yes / Partial	Yes / Yes	Mostly supported
Constructed oracle is no worse than routing	Partial / No	Partial / Partial	Mixed

7.6 Earlier distance-scaled implicit-head probe

All 1L–8L runs obey the 192-token native-operation limit; violations are zero. The largest call is 160 tokens at 1L and 192 thereafter. Table 9 gives mean and median over five seed means, each computed from four examples.

Table 9: Implicit-head answer loss. Each cell is mean / median over five seed means.

Tier	Position	1L	2L	4L	8L
Tiny	Absolute	.143/.104	.805/.920	1.338/1.382	1.206/1.403
Tiny	RoPE	1.043/.734	.543/.017	.830/.032	1.023/.565
Small	Absolute	.002/.002	1.520/.002	1.524/.004	1.618/.005
Small	RoPE	.001/.001	.062/.013	.129/.011	.282/.042

Small absolute means are dominated by one catastrophic seed; four of five seeds remain near zero. RoPE avoids that catastrophe but is lower in only two of five paired seeds at 2L–8L. Tiny results also change direction by seed and distance. This is reduced catastrophic variance in this sample, not a robust quality win. RoPE memory benefit over its own disabled condition is positive at 2L and 4L but negative at 8L: positional stability can coexist with unhelpful retrieval.

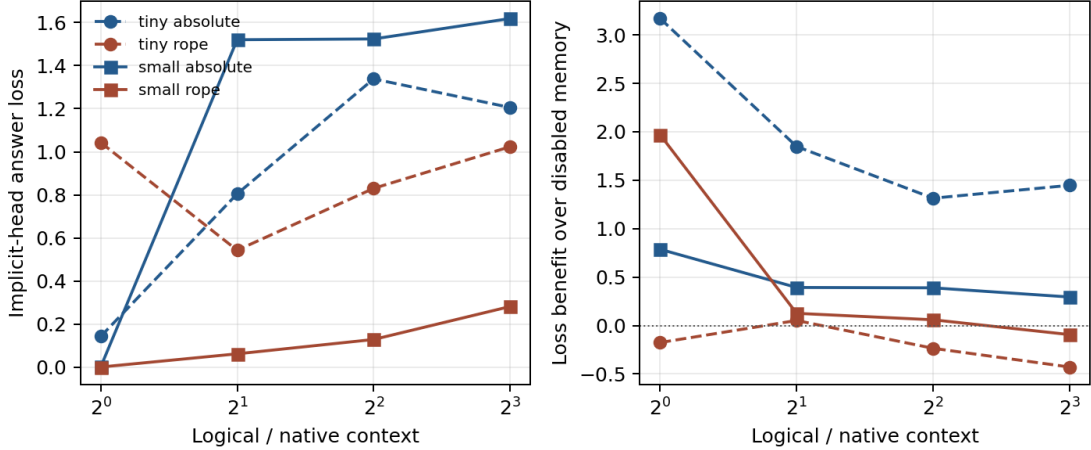


Figure 8: Bounded-head quality and benefit over disabled memory. Mean curves must be read with Table 9 because seed outliers are large.

8 Discussion

What the position contract solves. Source-relative offsets, not RoPE alone, restore first-layer coordinate identity. Learned absolute positions do so within their table capacity; RoPE decouples formula-defined coordinates from that table and provides common-translation semantics. The continuous-head experiment confirms that a logical span can advance while every native operation remains bounded.

What the contract does not solve. It does not restore left context lost by independent encoding, choose the relevant chunk, prevent harmful attention to irrelevant K/V, or guarantee extrapolation from short training contexts. The small-absolute overlap reversal, tiny-RoPE oracle reversal, small native-all reversal, and weak routed conditions are useful counterexamples: position correctness is necessary for continuity but insufficient for useful retrieval.

Post- versus pre-position K. Post-position K is immediately compatible with attention but binds the cache to source positions. Pre-position K is portable and compact offset metadata is small, but deferred rotation adds materialization work and requires a distance policy. Here it is 3.62 times the 160-token combined warm microbenchmark and does not reduce cache bytes. Exact logical assignment is faithful; compressed, bucketed, and order-preserving relocation change attention enough that their task quality remains an empirical question.

Engineering implication. Source identity, logical range, position state, and positional mechanism belong in cache provenance even when K is post-position. Without them, a consumer cannot audit local versus logical coordinates, reproduce a score, or safely introduce deferred rotation. For contiguous chunks, one logical start is enough for execution; full IDs can be reconstructed for traces.

9 Limitations and Threats to Validity

These are educational controlled models, not pretrained LLMs. The character tokenizer, binary task, 12 split examples, four head examples, and GTX 950M limit external validity. Five seeds permit paired direction checks but weak exact significance; an all-same-direction effect still has two-sided $p = 0.0625$. Mean loss is sensitive to rare catastrophes. The older split benchmark uses independent references with effective local positions and should not be read as the continuous-head source-offset result. The constructed oracle assumes the first evidence chunk and can omit useful context; it is a diagnostic intervention, not known optimal selection. Pre-position storage is not the canonical runtime path. We do not report production throughput, large-model memory savings, natural-language QA, or unrestricted streaming quality. Streaming tests establish bounded rollover mechanics for absolute and RoPE modes, not useful generation beyond the native horizon. The distance-policy probe uses random content and must not be read as a language-model accuracy comparison.

10 Reproducibility

The pre-change baseline is tagged `paper1-pre-rope-baseline`; development occurs on `research/paper1-5-rope`. The source-offset decomposition, storage matrix, continuous-head progression, and component timings use code revision `d36f67752701`; the structured artifacts record the full SHA. Earlier model training and split-quality measurements use `2b50f34cbf9f`. Raw JSON/CSV, plots, environment metadata, expectations, training histories, and manifests live under `docs/papers/shared/results/paper1.5_rope/`. Checkpoints are under `out/paper1.5_rope/`. Tests cover translation, norm preservation, large logical offsets through overlap and splitting, independent absolute operation/position bounds, cache metadata, exact no-memory SA-to-PRA conversion, policy order, causality, deferred rotation, streaming rollover, and CPU/GPU parity. Existing Paper 1 tests remain the absolute-mode guardrail.

11 Conclusion

Retrieved native-KV memory needs an explicit positional contract. A compact source offset removes first-layer reset error for learned absolute positions and RoPE alike. RoPE additionally supports deferred binding and clean common translation, but the unfused raw-K path is slower and K-only displacement deliberately changes attention. Deeper representations still depend on causal encoding context, and end-task quality still depends on selection and the composition of retrieved chunks. The next test is therefore not merely a longer coordinate range: it is pretrained-model evaluation that measures position identity, contextual recomputation, routing recall, and answer quality separately.

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